

SAVER: Additional Results

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1 Additional Results

Table 1: Probe-coverage sensitivity on CounterFact using Qwen2.5-7B + AlphaEdit. The selected online probe bank is varied from 25% to 100%, while evaluation uses a fixed disjoint held-out audit bank. Lower distortion risk is better, while higher NSR and monitored-held-out risk correlation ρ_D are better.

Selected probe coverage	Acc. $\uparrow$	Selected-set $\hat{D} \downarrow$	Held-out $\hat{D} \downarrow$	Held-out NSR $\uparrow$	$\rho_D \uparrow$	$\hat{\beta}_T$
25%	0.977	0.088	0.121	0.142	0.421	0.880
50%	0.982	0.093	0.109	0.150	0.548	0.850
75%	0.981	0.091	0.103	0.152	0.603	0.860
100%	0.986	0.094	0.095	0.157	0.667	0.835

Table 2: Matched-acceptance comparison with heuristic admission gates on CounterFact using Qwen2.5-7B + AlphaEdit. All-request ESR and PSR count rejected requests as failures. Current-probe-loss, KL-drift, and random-rejection controls are matched to SAVER’s acceptance rate.

Method	Acc. $\uparrow$	ESR $\uparrow$	PSR $\uparrow$	NSR $\uparrow$	PPL $\downarrow$	$\hat{D} \downarrow$
SAVER	0.986	0.866	0.618	0.157	41.67	0.095
Current-probe-loss gate	0.985	0.847	0.599	0.142	45.36	0.116
KL-drift gate	0.987	0.841	0.592	0.148	44.52	0.110
Random rejection	0.986	0.831	0.582	0.127	49.12	0.142

For Table 2, we analyze them at matched SAVER acceptance rate with following conditions:

- Current-probe-loss gate: Accept the edit only if it does not increase loss on the current online probes beyond the chosen threshold.
- KL-drift gate: Accept the edit only if the model’s output distribution on probes does not drift too far after the edit.
- Random rejection: Accept the same fraction of edits as SAVER, but choose which edits to reject randomly.

Table 3: Long-sequence analysis on CounterFact using Qwen3.5-8B + AlphaEdit. Results are reported at $T \in \{100, 500, 1000, 2500, 5000\}$ submitted edits.

Submitted edits T	Method	Acc. $\uparrow$	ESR $\uparrow$	PSR $\uparrow$	NSR $\uparrow$	PPL $\downarrow$	$\widehat{D} \downarrow$	$\widehat{\beta}_T$
100	Unconstrained	1.000	0.886	0.640	0.190	34.70	0.108	–
100	+SAVER	0.992	0.901	0.653	0.205	33.50	0.087	0.820
500	Unconstrained	1.000	0.851	0.602	0.133	47.90	0.144	–
500	+SAVER	0.987	0.872	0.624	0.163	40.80	0.094	0.840
1,000	Unconstrained	1.000	0.801	0.552	0.104	64.30	0.181	–
1,000	+SAVER	0.979	0.848	0.598	0.151	47.20	0.093	0.835
2,500	Unconstrained	1.000	0.695	0.455	0.069	103.50	0.271	–
2,500	+SAVER	0.967	0.792	0.544	0.124	59.80	0.098	0.860
5,000	Unconstrained	1.000	0.574	0.357	0.044	162.70	0.348	–
5,000	+SAVER	0.952	0.731	0.481	0.102	76.40	0.097	0.855

Table 4: Probe-quality analysis on CounterFact using Qwen2.5-7B + AlphaEdit. The standard in-scope generality and locality probes are compared with a semantically unrelated probe set of the same size.

T	Probe set type	Acc. $\uparrow$	$\widehat{D}_{\text{sel}} \downarrow$	$\widehat{D}_{\text{hold}} \downarrow$	NSR _{hold} $\uparrow$	$\rho_D \uparrow$	$\widehat{\beta}_T$
100	In-scope	0.989	0.096	0.101	0.179	0.592	0.825
100	Semantically unrelated	0.991	0.089	0.116	0.163	0.322	0.775
250	In-scope	0.987	0.093	0.097	0.166	0.641	0.840
250	Semantically unrelated	0.990	0.090	0.119	0.146	0.281	0.785
500	In-scope generality + locality	0.986	0.092	0.095	0.157	0.660	0.835
500	Semantically unrelated	0.991	0.086	0.124	0.140	0.274	0.780

Table 5: Controlled persistent-risk detection. The monitor is initialized with a 100-round below-threshold phase and then subjected to persistent post-change risk. We use $\theta = 0.10$, $\alpha = 0.10$, full evaluation $q_t = 1$, the fixed boundary $\beta = 0.99$, and the 23-point grid $G = \{0.55, 0.57, \dots, 0.97, 0.99\}$. Results are summarized over 2,000 replications with a 2,000-round post-change horizon. Delay is measured from the change point to threshold crossing.

Post-change Risk	Excess Risk	Fixed- β Delay Median / P90	Grid Delay Median / P90	Pre-detection Excess-risk Commits Median	Detected Within 2,000 Rounds
0.137	0.037	400 / 560	535 / 715	534	$\geq 99.5\%$
0.236	0.136	100 / 122	123 / 147	122	100%
0.333	0.233	64 / 75	78 / 89	77	100%

Table 6: Boundary-grid specification and prediction-set sizes. We use the 23-point grid $G = \{0.55, 0.57, \dots, 0.97, 0.99\}$, with $\beta_{\text{max}} = 0.99 < 1$. The reported average operating level $\widehat{\beta} \approx 0.835$ is not an individual grid value; its neighboring selectable boundaries are 0.83 and 0.85. The 0.99 rows show the capped upper-endpoint reference, while SAVER selects the smallest admissible boundary through Eq. (14).

Dataset	Evaluation Level	Mean Set Size	Median	P90
CounterFact	Grid boundary $\beta = 0.83$	266.3	126	592
	Mean operating level $\beta = 0.835$ (post hoc)	282.2	135	629
	Grid boundary $\beta = 0.85$	337.6	163	764
	Upper endpoint $\beta = 0.99$	8830.3	8097	15721
zsRE	Grid boundary $\beta = 0.83$	150.2	72	364
	Mean operating level $\beta = 0.835$ (post hoc)	160.1	77	388
	Grid boundary $\beta = 0.85$	195.3	96	474
	Upper endpoint $\beta = 0.99$	7475.4	6280	14434